

The Aggregate Opportunity Cost of Recreational Land Use: A Counterfactual Valuation of U.S. Golf Courses

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Abstract

This thesis investigates the macroeconomic inefficiency of recreational land allocation in the United States, specifically quantifying the opportunity cost of maintaining golf courses in supply-constrained markets. Utilizing a novel spatial dataset of 16,297 facilities, this study applies the real estate appraisal principle of Highest and Best Use (HBU) to establish residential and agricultural counterfactuals. The methodology integrates Federal Housing Finance Agency (FHFA) and United States Department of Agriculture (USDA) land valuations, employing a rural-urban continuum bifurcation to control for spatial elasticity. To rigorously resolve 28.8% missing acreage data, the analysis utilizes Multiple Imputation by Chained Equations (MICE) with a Random Forest backend and pools the parameters using Rubin's Rules. Empirical results, cross-validated across Python, R, and Julia pipelines, reveal a pooled aggregate opportunity cost of approximately **\$943 billion (95% CI: \$936B – \$950B)**. This locked land value is characterized by extreme spatial heterogeneity; while urban courses comprise 70% of the total facilities, they account for over 98% of the aggregate financial opportunity cost, reflecting the massive regulatory “zoning tax” endemic to metropolitan areas. A pooled logarithmic regression confirms this, indicating that an urban location increases a course's opportunity cost by a factor of approximately 60. Despite this immense latent value, the market fails to transition these parcels to their HBU due to severe institutional frictions. A dual layer of exclusionary municipal zoning and inflexible private restrictive covenants effectively paralyzes land-use transitions, rendering Coasian bargaining impossible. These findings highlight a systemic market failure, suggesting that unlocking this land for high-density housing or utility-scale renewable energy infrastructure requires aggressive legal and regulatory reform.

1 Introduction

The allocation and regulation of urban and suburban land represent critical challenges within modern spatial economics. In many major United States metropolitan areas, the supply of developable land is highly inelastic, driven largely by restrictive land-use regulations rather than physical geographic constraints (E. Glaeser et al. 2017; Gyourko et al. 2021). This artificial scarcity has led to a pronounced divergence between the physical cost of construction and the market price of housing, generating a substantial “zoning tax” that inflates residential land values (E. L. Glaeser et al. 2005; Davis, Larson, et al. 2021).

Amidst this intense competition for spatial resources, the American golf course represents a massive and highly specific land-use typology. Historically, real estate developers utilized golf courses as “surrogate public goods” to cross-subsidize the construction of master-planned communities, capturing exorbitant price premiums on adjacent residential lots (Frank F Limehouse et al. 2012). However, empirical evidence suggests that the marginal amenity value of proximity to golf courses has depreciated over time as consumer preferences shift toward natural open spaces (Cho et al. 2009). Despite this structural decline in recreational demand and the escalating value of residential land, hundreds of thousands of acres of prime real estate remain locked in this single-use capacity (Weinand et al. 2025).

To quantify the macroeconomic inefficiency of this spatial allocation, this thesis applies the foundational real estate appraisal principle of Highest and Best Use (HBU). HBU dictates that land should be valued based on its most profitable, legally permissible, and physically possible use (Appraisal Institute 2020). By treating the existing golf course as a baseline, we can establish a counterfactual valuation model to determine the aggregate opportunity cost (V_{OC}) of this land. The core economic framework is defined by the following equation:

$$V_{OC} = \max(V_{Res}, V_{Ag}) - V_{Current} \quad (1)$$

where V_{Res} represents the land value under a residential development counterfactual, V_{Ag} represents the value under an agricultural counterfactual, and $V_{Current}$ represents the capitalized value of the land in its current recreational use.

In a frictionless, perfectly competitive market, land would naturally transition to the use that maximizes its residual value. However, the transition of golf course land to its HBU is severely impeded by a dual layer of institutional frictions. Publicly, municipal zoning ordinances frequently prohibit high-density redevelopment. Privately, properties are often encumbered by stale restrictive covenants and Homeowner Association (HOA) agreements that legally mandate the preservation of the golf course in perpetuity, regardless of its financial feasibility (Ellickson 2022; Korngold 2024).

Because these legal barriers prevent fluid Coasian bargaining and market optimization, this thesis employs a formal *Hypothetical Condition* (Appraisal Institute 2020). The analysis proceeds under the assumption of a frictionless legal environment —specifically, the hypothetical dissolution of both exclusionary zoning restrictions and parallel private covenants. This methodological assumption is necessary to isolate and measure the pure economic opportunity cost of the land.

Therefore, this thesis seeks to answer the following core research question:

What is the aggregate financial opportunity cost of U.S. golf courses when evaluated against their Highest and Best Use counterfactuals of residential, urban, or agricultural development?

To answer this question, this study constructs a national-level spatial dataset integrating golf course boundaries from OpenStreetMap (OSM), paired with granular land valuation data from the Federal Housing Finance Agency (FHFA) and the United States Department of Agriculture (USDA). To address missing spatial covariates, the analysis utilizes Multiple Imputation by Chained Equations (MICE), ensuring that the final aggregate valuation remains statistically robust and accounts for between-imputation variance (Buuren 2018; Rubin 1987). Ultimately, by quantifying this locked land value, this research aims to inform broader policy discussions regarding zoning reform, housing supply, and the economic efficiency of urban land use.

1.1 Roadmap

The remainder of this thesis is organized as follows:

Section 2 reviews the relevant economic and legal literature.

Section 3 formalizes the theoretical framework, translating HBU into a neoclassical spatial model.

Section 4 details the data and methodology.

Section 5 presents the empirical results, reporting the pooled aggregate valuation of **\$943 billion**, decomposing this cost, and providing the pooled logarithmic regression outputs.

Section 6 discusses the macroeconomic and policy implications of these findings, including a micro-case study of Hawaii.

Finally, **Section 7** concludes the thesis with a summary and directions for future economic research.

2 Literature Review

This thesis bridges three distinct strands of economic and legal literature: the macroeconomic valuation of urban land and regulatory scarcity, the microeconomic pricing of recreational amenities, and the institutional frictions that govern land-use transitions. By synthesizing these domains, this research establishes the theoretical foundation necessary to quantify the aggregate opportunity cost of golf course land.

2.1 Urban Land Valuation and Regulatory Scarcity

The primary driver of the opportunity cost associated with underutilized urban land is the escalating value of residential real estate (V_{Res}). Foundational work by Davis and Palumbo (2008) decomposes housing prices into structure costs and land values, demonstrating that the vast heterogeneity in U.S. housing prices is driven almost entirely by the price of the underlying land, rather than physical construction costs. Davis, Larson, et al. (2021) expand upon this by providing highly granular, tract-level estimates of residential land prices, confirming that land values follow steep spatial gradients that peak sharply near central business districts.

This extreme appreciation in land value is largely attributed to artificial supply constraints. E. L. Glaeser et al. (2005) introduce the concept of the “zoning tax,” defined as the gap between the market price of a housing unit and its marginal cost of production. In a frictionless market, free entry ensures that prices equal average costs; however, restrictive land-use regulations sever this relationship, bidding up the extensive margin value of land (Gyourko et al. 2021). The macroeconomic implications of this inelastic housing supply are severe. E. Glaeser et al. (2017) argue that regulatory constraints in high-productivity markets prevent the efficient spatial allocation of labor, resulting in significant deadweight loss to aggregate national output. Consequently, the opportunity cost of maintaining large tracts of single-use recreational land in these constrained markets is not merely a localized inefficiency, but a macroeconomic distortion.

2.2 The Microeconomics of Recreational Amenities

To accurately model the current value of golf courses ($V_{Current}$), it is necessary to understand the economic mechanisms that generate their revenue. Utilizing the two-stage hedonic pricing model established by Rosen (1974), Frank F. Limehouse et al. (2010) demonstrate that environmental quality and recreational exclusivity are credence goods capitalized directly into golf course pricing.

However, the primary economic driver of golf course development is often not the recreational operation itself, but the “joint production” of the course and adjacent real estate (Frank F. Limehouse et al. 2012). Developers utilize the golf course as a surrogate public good to cross-subsidize the project, extracting massive price premiums from the surrounding residential lots. While this model historically maximized developer profit, longitudinal hedonic analyses indicate that the amenity value of golf courses is subject to structural depreciation. Cho et al. (2009) find that while proximity to open space acts as a substitute for private lot size, consumer preferences have shifted significantly over time. The marginal value of proximity to natural amenities (e.g., public parks, water bodies) has increased, whereas the premium associated specifically with golf courses has experienced a measurable decline. Furthermore, emerging literature by Weinand et al. (2025) highlights the severe spatial inefficiency of golf courses, noting that the land footprint of these facilities vastly exceeds the land required for utility-scale renewable energy infrastructure, introducing alternative, non-residential counterfactuals for land optimization.

2.3 Institutional Frictions and Land Use Rigidity

If the residential value of land (V_{Res}) exceeds its current recreational value ($V_{Current}$), standard neoclassical theory dictates that the land should be redeveloped. The failure of the market to execute this transition is explained by severe institutional frictions and transaction costs, which prevent fluid Coasian bargaining (Coase 2013).

The rigidity of current land use is enforced by a dual layer of public and private regulation. Publicly, municipal zoning laws frequently prohibit the densification required to achieve Highest and Best Use (Fischel

2015). While these planning policies generate localized environmental amenities, Cheshire and Sheppard (2002) demonstrate that the net welfare effect of such restrictive regulatory regimes is substantially negative and acts as a regressive tax on housing affordability.

Privately, land-use transitions are blocked by restrictive covenants. Ellickson (2022) notes that the majority of modern housing developments are tethered by covenants that mandate the preservation of existing land uses (such as golf courses) in perpetuity. Because these covenants typically require supermajority votes to amend, they suffer from severe collective action problems and status quo bias, effectively freezing the land use. To resolve this, Korngold (2024) argues that legislative zoning reform is insufficient unless parallel private covenants are simultaneously voided under public policy doctrines. This literature confirms that the legal barriers to redevelopment are formidable, thereby justifying this study’s use of a Hypothetical Condition to calculate the pure economic opportunity cost absent these specific legal frictions.

2.4 Synthesis and Contribution

While existing literature extensively documents the regulatory inflation of residential land values and the microeconomic pricing of golf course amenities, there is a distinct lack of empirical research quantifying the aggregate spatial intersection of these two phenomena. This thesis contributes to the literature by applying rigorous real estate appraisal frameworks (Appraisal Institute 2020) to granular spatial datasets, calculating the precise national opportunity cost of maintaining depreciating recreational assets in supply-constrained housing markets.

3 Theoretical Framework

This section establishes the formal economic model utilized to quantify the spatial misallocation of recreational land. The framework integrates neoclassical urban economics with the professional appraisal doctrine of Highest and Best Use (HBU) to derive a mathematical function for aggregate opportunity cost.

3.1 Neoclassical Land Allocation and Highest and Best Use

In a frictionless, perfectly competitive spatial market, land is allocated to the use that generates the maximum residual value. This principle aligns with the appraisal standard of Highest and Best Use (HBU), defined as the reasonably probable and legal use of vacant land or an improved property that is physically possible, appropriately supported, financially feasible, and results in the highest value (Appraisal Institute 2020).

For a given parcel of land i , let $V_{i,k}$ represent the present value of the land under use k , where $k \in \{1, 2, \dots, K\}$ represents the set of all physically possible and financially feasible uses. Under conditions of spatial equilibrium, the market value of the parcel, V_i^* , is determined by the use that maximizes this value:

$$V_i^* = \max\{V_{i,1}, V_{i,2}, \dots, V_{i,K}\} \quad (2)$$

When the current use of the land, $V_{Current}$, diverges from V_i^* , an economic inefficiency exists. The magnitude of this inefficiency is the opportunity cost.

3.2 The Opportunity Cost Function

To quantify the inefficiency of land currently allocated to golf courses, this analysis restricts the set of counterfactual uses K to the two most probable alternative land uses based on spatial geography: residential development (V_{Res}) and agricultural production (V_{Ag}).

The fundamental opportunity cost (V_{OC}) for a specific golf course parcel i is defined as the difference between its value under its HBU counterfactual and its value in its current recreational state:

$$V_{OC,i} = \max(V_{Res,i}, V_{Ag,i}) - V_{Current,i} \quad (3)$$

The aggregate national opportunity cost is the summation of $V_{OC,i}$ across all N golf courses in the dataset:

$$\text{Aggregate } V_{OC} = \sum_{i=1}^N [\max(V_{Res,i}, V_{Ag,i}) - V_{Current,i}] \quad (4)$$

3.2.1 Deconstructing the Counterfactuals

The variables within the core equation are defined by distinct microeconomic determinants:

- **V_{Res} (Residential Value):** The extensive margin value of the land if subdivided and developed for single-family or multi-family housing. In highly regulated metropolitan statistical areas (MSAs), V_{Res} is artificially inflated by a regulatory “zoning tax” (τ_z), which represents the scarcity premium induced by supply-side restrictions (E. L. Glaeser et al. 2005; Gyourko et al. 2021).
- **V_{Ag} (Agricultural Value):** The capitalized value of the land’s agricultural yield, primarily applicable to courses located in rural or exurban geographies where residential demand is insufficient to justify development costs.
- **$V_{Current}$ (Current Recreational Value):** The capitalized net operating income of the golf course. This value is heavily dependent on the joint production model, where the course serves as a surrogate public good that generates price premiums for adjacent real estate (Frank F Limehouse et al. 2012).

3.3 Institutional Frictions and Market Failure

If $V_{OC,i} > 0$, rational economic actors should execute a transaction to redevelop the land. The persistence of golf courses in high-value urban areas indicates a market failure driven by severe institutional frictions (F).

These frictions consist of public regulatory constraints (Z) and private contractual constraints (C) (Ellickson 2022; Korngold 2024). Redevelopment imposes a negative externality on adjacent homeowners by destroying their capitalized open-space amenity, leading to intense political resistance and prohibitive Coasian transaction costs (Coase 2013).

Mathematically, redevelopment only occurs if the opportunity cost exceeds the total institutional frictions:

$$V_{OC,i} > F(Z_i, C_i) \tag{5}$$

Because $F(Z_i, C_i)$ can be insurmountable, the market cannot clear. To isolate the pure economic value of the land, this thesis applies a *Hypothetical Condition*, setting $F(Z_i, C_i) = 0$. This allows for the unconstrained calculation of V_{OC} .

4 Data and Methodology

To quantify the aggregate opportunity cost of U.S. golf courses, this study constructs a novel, national-level spatial dataset. The methodology integrates geospatial boundary manipulation, a hybrid valuation algorithm, and advanced imputation techniques, with the entire pipeline cross-validated in Python, R, and Julia to ensure robustness.

4.1 Dataset Scope and Sourcing

The primary dataset consists of $N = 16,297$ golf course observations. Spatial point coordinates (latitude and longitude) for each facility were spatially joined to U.S. Census Bureau County shapefiles (2022 vintage) to assign a 5-digit Federal Information Processing Standard (FIPS) code to each observation.

To establish the HBU counterfactuals, two distinct 2022 land valuation datasets were integrated:

1. **Residential Land Values (V_{Res}):** Sourced from the Federal Housing Finance Agency (FHFA) 2022 county-level panel data.
2. **Agricultural Land Values (V_{Ag}):** Sourced from the USDA National Agricultural Statistics Service (NASS) 2022 county-level survey data.

4.2 The Hybrid Valuation Algorithm

A hybrid valuation algorithm was developed utilizing the 2023 USDA Rural-Urban Continuum Codes (RUCC). Observations were bifurcated based on their RUCC designation:

- **Urban Classification (RUCC 1, 2, 3):** Courses in metropolitan counties were assigned the FHFA county-level residential price per acre (V_{Res}).
- **Rural Classification (RUCC 4 through 9):** Courses in non-metropolitan counties were assigned the USDA county-level agricultural price per acre (V_{Ag}).

This process resulted in a '*Baseline_{ValuePerAcre}*' for approximately 93.3% of the courses.

4.3 Handling Missing Covariates: Acreage and Baseline Value

A critical limitation was the absence of acreage data for all courses and missing baseline values for 6.7% of courses.

- **Acreage Extraction:** A three-tier strategy was employed. First, physical boundaries for 15,166 golf courses were extracted from an 11 GB OpenStreetMap (OSM) '.pbf' file using a memory-efficient 'pyosmium' stream. A two-pass spatial join matched these polygons to 71.2% of the course list. For the remaining misses, a fallback join was attempted against the US Census 'tigris' "Area Landmarks" dataset. A new 'acreage_source' column tracks the provenance of each value ('OSM', 'Tigris', or 'MICE_Target').
- **Multiple Imputation:** The remaining missing 'final_acreage' and 'Baseline_Value_Per_Acre' were imputed using Multiple Imputation by Chained Equations (MICE). Following Rubin (1987), $m = 5$ parallel, complete datasets were generated. The imputation algorithm utilized Random Forest-based methods ('miceforest' in Python, 'mice' with 'method='rf'' in R) to handle non-linear spatial relationships and prevent impossible negative value predictions, a known risk with linear imputers (Buuren 2018). The predictor matrix included 'Holes', 'Ownership_Type', 'county_type', 'Longitude', and 'Latitude'.

4.4 Parameter Pooling and Rubin's Rules

All final statistical estimates were derived by performing the analysis independently across the $m = 5$ imputed datasets and then pooling the results using Rubin's Rules. The aggregate national land value was calculated by summing the total opportunity cost within each of the five datasets and then taking the arithmetic mean

of those five sums. This process ensures the final aggregate figure and its confidence interval accurately reflect the underlying uncertainty introduced by the imputation process (Buuren [2018](#)).

5 Empirical Results

This section presents the quantitative findings derived from the spatial valuation model and the multiply imputed datasets. The results quantify the aggregate opportunity cost of golf course land and decompose this value across spatial and operational dimensions.

5.1 Aggregate Opportunity Cost and Imputation Stability

The primary objective of this analysis is to quantify the aggregate opportunity cost (V_{OC}) of the 16,297 golf courses. Utilizing the hybrid valuation algorithm and pooling the results across the $m = 5$ imputed datasets via Rubin’s Rules, the aggregate baseline land value is estimated at **\$943.0 billion**.

The standard error of the pooled estimate is \$3.43 billion, yielding a 95% confidence interval of **\$936.3 billion to \$949.7 billion**. The narrow confidence interval relative to the aggregate sum (a spread of only 1.4%) indicates high statistical stability, confirming that the between-imputation variance introduced by the missing acreage data does not compromise the macroeconomic significance of the finding.

5.2 Spatial Heterogeneity and Value Concentration

The aggregate opportunity cost is not distributed uniformly across the spatial landscape. Table 1 details the bifurcation of the dataset into Urban and Rural classifications, based on the pooled results.

Table 1: Pooled Spatial Bifurcation of Golf Course Opportunity Costs

County Type	Observations (N)	Total Acreage	Total Est. Value	Avg. Price Per Acre
Urban	11,391	1,635,000	\$929.0 Billion	\$568,195
Rural	4,872	665,000	\$14.0 Billion	\$21,052
Unclassified	34	4,500	—	—
Total	16,297	2,304,500	\$943.0 Billion	—

The data reveal an extreme spatial concentration of value. While Urban courses account for 69.9% of the total facilities and 71.0% of the total acreage, they represent ****98.5%**** of the total aggregate opportunity cost. The average price per acre for an Urban course (\$568,195) is over 27 times higher than that of a Rural course (\$21,052).

This finding provides robust empirical confirmation of the theoretical frameworks established by Davis, Larson, et al. (2021) and E. L. Glaeser et al. (2005). The massive opportunity cost of golf course land is a distinctly urban and suburban phenomenon, driven by the “zoning tax” and artificial land scarcity in highly regulated metropolitan statistical areas.

5.3 Econometric Determinants of Land Value

To further analyze the drivers of this opportunity cost, a pooled logarithmic regression was estimated across the $m = 5$ datasets. The model regresses the natural log of the total opportunity cost against physical and operational covariates. The pooled results are presented in Table 2.

All covariates in the model are statistically significant at the $p < 0.001$ level. The coefficient for ‘Holes’ ($\beta = 0.047$) indicates that each additional hole increases the underlying opportunity cost by approximately 4.8% ($e^{0.047} - 1$), holding other factors constant.

The spatial coefficient ‘*countytype : Urban*’ ($\beta = 4.172$) is the most powerful predictor. Controlling for course size and type, locating a golf course in an urban county increases its opportunity cost by a factor of approximately ****64.8**** ($e^{4.172}$). This mathematically isolates the immense magnitude of the regulatory scarcity premium (τ_z) that prevents this land from transitioning to its Highest and Best Use.

Table 2: Pooled Regression Output: $\ln(\text{Total_Opportunity_Cost})$

Term	Estimate (β)	Std. Error	t-statistic	p-value
(Intercept)	12.282	0.039	318.55	< 0.001
Holes	0.047	0.002	20.06	< 0.001
Ownership_Type: Private	0.205	0.028	7.28	< 0.001
Ownership_Type: Public	-0.214	0.025	-8.45	< 0.001
county_type: Urban	4.172	0.023	184.80	< 0.001

Note: Results from the Python pipeline. Reference categories are Municipal for Ownership_Type and Rural for county_type. Standard errors are heteroskedasticity-robust (HC1).

6 Discussion and Policy Implications

The empirical results demonstrate that the aggregate opportunity cost of maintaining golf courses in the United States is approximately **\$943 billion**. Crucially, 98.5% of this foregone value is concentrated in urban and suburban counties. This section contextualizes these findings, examines the institutional frictions preventing market optimization, and explores alternative land-use counterfactuals.

6.1 The Magnitude of Spatial Misallocation

The **\$929 billion** opportunity cost isolated within urban counties represents a massive spatial misallocation of capital. In highly regulated, inelastic housing markets, the restriction of land supply artificially inflates residential land values (V_{Res}) (E. L. Glaeser et al. 2005; Davis, Larson, et al. 2021). By dedicating over 1.6 million acres of prime urban real estate to a single-use recreational amenity, municipalities are implicitly accepting a severe reduction in potential housing density.

As E. Glaeser et al. (2017) note, the inability to deliver housing in high-productivity urban centers restricts labor mobility and depresses aggregate national economic output. Therefore, the \$929 billion figure is not merely a localized appraisal metric; it is a quantification of the deadweight loss imposed by maintaining low-density recreational land in areas experiencing acute housing shortages.

6.2 Institutional Frictions and the Failure of Coasian Bargaining

The failure of the market to redevelop this land highlights severe institutional frictions that render Coasian bargaining impossible (Coase 2013). The primary friction is the capitalization of the golf course amenity into surrounding real estate, which incentivizes fierce political resistance (NIMBYism) from homeowners protecting their asset values (Cheshire and Sheppard 2002).

Furthermore, even if municipal zoning is amended, private contractual law—specifically restrictive covenants enforced by HOAs—frequently supersedes public policy (Ellickson 2022). Because these covenants require supermajority approval to amend, they suffer from insurmountable collective action problems, effectively freezing the land use in perpetuity. To resolve this paralysis, Korngold (2024) proposes that state legislatures or judiciaries must void these “stale” covenants under the doctrine of violating public policy. However, this carries significant constitutional risk, as voiding a private covenant may be classified as a regulatory taking requiring “just compensation.” Consequently, the \$943 billion opportunity cost remains locked not by a lack of market demand, but by an intractable web of property law and transaction costs.

6.3 Micro-Case Study: The Valuation Gap in Hawaii

To ground the macroeconomic model in real-world policy, a micro-case study was conducted on the 33 distinct golf courses in Honolulu County, Hawaii. The analysis revealed a significant “Unit of Analysis Mismatch” between functional golf courses and their underlying legal tax parcels (Tax Map Keys, or TMKs). A spatial intersection identified 1,072 unique TMKs constituting the 33 courses.

The study pivoted from a direct financial comparison to a geographic fragmentation analysis. The key finding is that **63.2% of all golf course parcels** are concentrated in the Ewa district (Zone 9), the primary vector for new housing development and urban expansion on the island. This provides direct empirical evidence that golf courses are disproportionately occupying the exact land most critical for alleviating the region’s severe housing shortage. This finding strongly suggests that the official assessed values, based on current recreational use, systematically understate the land’s true economic potential under its HBU as residential development.

6.4 Alternative Counterfactuals: The Renewable Energy Imperative

Given the severe frictions associated with residential redevelopment, policymakers must consider alternative HBU counterfactuals. Weinand et al. (2025) identify utility-scale renewable energy infrastructure as a highly synergistic alternative. The spatial footprint of U.S. golf courses is approximately four times larger than the total land area currently dedicated to utility-scale solar installations. Future policy frameworks should integrate $V_{Renewable}$ as a primary counterfactual, particularly in arid urban environments where the ecological cost of maintaining turf exacerbates the economic opportunity cost.

6.5 Methodological Limitations

The primary limitation of this study is the use of a marginal land value for a bulk parcel valuation. The FHFA data reflects the per-acre price of existing, subdivided residential lots. In professional appraisal practice, a developer acquiring a bulk 150-acre parcel receives a discount for subdivision costs, infrastructure development, and absorption risk (Appraisal Institute 2020). A sensitivity analysis applying a conservative 25% bulk discount to all urban parcels still yields a national opportunity cost exceeding \$700 billion, confirming the macroeconomic significance of the finding. Therefore, the \$943 billion figure should be interpreted as the *gross theoretical opportunity cost* of the land, rather than its immediate bulk liquidation value.

7 Conclusion

This thesis quantified the aggregate opportunity cost of maintaining 16,297 U.S. golf courses, establishing a robust empirical estimate of approximately **\$943 billion**. This spatial misallocation is not uniformly distributed; 98.5% of this value is concentrated in metropolitan areas, driven by the regulatory “zoning tax” that artificially inflates residential land values (E. L. Glaeser et al. 2005; Davis, Larson, et al. 2021).

The market fails to transition these parcels to their HBU not due to a lack of demand, but due to severe institutional frictions. A dual layer of public zoning and private restrictive covenants paralyzes land-use transitions, rendering Coasian bargaining impossible and legally barring redevelopment (Coase 2013; Ellickson 2022; Korngold 2024).

Consequently, unlocking this latent land value requires more than market forces; it necessitates aggressive legal and regulatory reform. Future economic research must expand upon this baseline valuation by modeling the specific transaction costs of voiding private covenants and by applying more sophisticated, market-specific plattage adjustments. Furthermore, as municipalities grapple with both housing shortages and the global energy transition, future spatial models should integrate utility-scale renewable energy as a primary, non-residential HBU counterfactual (Weinand et al. 2025). Ultimately, the \$943 billion opportunity cost quantified in this study serves as a stark financial metric of the deadweight loss imposed by inflexible land-use policies in the modern American economy.

References

- Appraisal Institute (2020). *The Appraisal of Real Estate*. 15th. Chicago, IL: Appraisal Institute. ISBN: 978-1935328780.
- Buuren, Stef van (2018). *Flexible imputation of missing data*. eng. Second edition. International and comparative criminal justice series. Boca Raton, FL: Chapman, Hall/CRC, an imprint of Taylor, and Francis. ISBN: 0-429-96035-2.
- Cheshire, Paul and Stephen Sheppard (2002). “The welfare economics of land use planning”. eng. In: *Journal of urban economics* 52.2, pp. 242–269. ISSN: 0094-1190.
- Cho, Seong-Hoon et al. (2009). “Spatial and temporal variation in the housing market values of lot size and open space”. eng. In: *Land economics* 85.1, pp. 51–73. ISSN: 0023-7639.
- Coase, R. H. (2013). “The Problem of Social Cost”. eng. In: *The Journal of law & economics* 56.4, pp. 837–877. ISSN: 0022-2186.
- Davis, Morris A., William D. Larson, et al. (2021). “The price of residential land for counties, ZIP codes, and census tracts in the United States”. eng. In: *Journal of monetary economics* 118, pp. 413–431. ISSN: 0304-3932.
- Davis, Morris A. and Michael G. Palumbo (2008). “The price of residential land in large US cities”. eng. In: *Journal of urban economics* 63.1, pp. 352–384. ISSN: 0094-1190.
- Ellickson, Robert C (2022). “STALE REAL ESTATE COVENANTS”. eng. In: *William and Mary law review* 63.6, pp. 1831–. ISSN: 0043-5589.
- Fischel, William A. (2015). *Zoning rules : the economics of land use regulation*. eng. 1st ed. Cambridge, Massachusetts: Lincoln Institute of Land Policy. ISBN: 1-55844-329-0.
- Glaeser, Edward, National Bureau of Economic Research, and Joseph Gyourko (2017). *The Economic Implications of Housing Supply*. eng. NBER working paper series no. w23833. Cambridge, Mass: National Bureau of Economic Research.
- Glaeser, Edward L., Joseph Gyourko, and Raven Saks (2005). “Why Is Manhattan So Expensive? Regulation and the Rise in Housing Prices”. eng. In: *The Journal of law & economics* 48.2, pp. 331–369. ISSN: 0022-2186.
- Gyourko, Joseph, National Bureau of Economic Research, and Jacob Krimmel (2021). *The Impact of Local Residential Land Use Restrictions on Land Values Across and Within Single Family Housing Markets*. eng. NBER working paper series no. w28993. Cambridge, Mass: National Bureau of Economic Research.
- Korngold, Gerald (2024). “Repealing Single-Family Zoning Is Not Enough: A Proposal For Removing Existing Parallel Private Covenants For Violating Public Policy”. eng. In: *Missouri law review* 89.1, pp. 1–. ISSN: 0026-6604.
- Limehouse, Frank F, Robert E McCormick, and Melissa M Yeoh (2012). “The private provision of public goods: an analysis of homes on golf courses”. eng. In: *The Journal of private enterprise* 27.2, pp. 103–120. ISSN: 0890-913X.
- Limehouse, Frank F., Peter C. Melvin, and Robert E. McCormick (2010). “The Demand for Environmental Quality: An Application of Hedonic Pricing in Golf”. eng. In: *Journal of sports economics* 11.3, pp. 261–286. ISSN: 1527-0025.
- Rosen, Sherwin (1974). “Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition”. eng. In: *The Journal of political economy* 82.1, pp. 34–55. ISSN: 0022-3808.
- Rubin, Donald B. (1987). *Multiple imputation for nonresponse in surveys*. eng. Wiley series in probability and mathematical statistics. Applied probability and statistics. New York ; Wiley. ISBN: 047108705X.
- Weinand, Jann Michael et al. (2025). “Countries across the world use more land for golf courses than wind or solar energy”. eng. In: *Environmental Research Communications* 7.2, pp. 21012–. ISSN: 2515-7620.